Printer & Scanner Basics

Printers

1. Introduction to Printers

Printers are commonly used output devices that produce a hard copy of document stored in electronic form, i.e they put information from computer on to paper.

There are various kinds of printers available today like Impact printers, Bubble-jet printers, Laser printers, Thermal printers etc.

2. Types of Printers

2.1 Impact Printers

Impact printers are among the old printing technologies, which make use of inked ribbon to make an imprint on the paper. Impact printers are considered noisy when compared to other printers.

The most commonly known impact printers are: *Daisy-Wheel Printers* and *Dot-Matrix Printers*.

2.1.1 Daisy-Wheel Printers

A Daisy-Wheel Printer works on the same principle as ball-head typewriter, using a disk with characters on the outer edge and a hammer (solenoid) to strike an ink ribbon and paper. Their speed is rated by cps (number of characters per second).

Daisy-wheel

A section in detail

Advantages: not expensive and can produce letter-quality text;

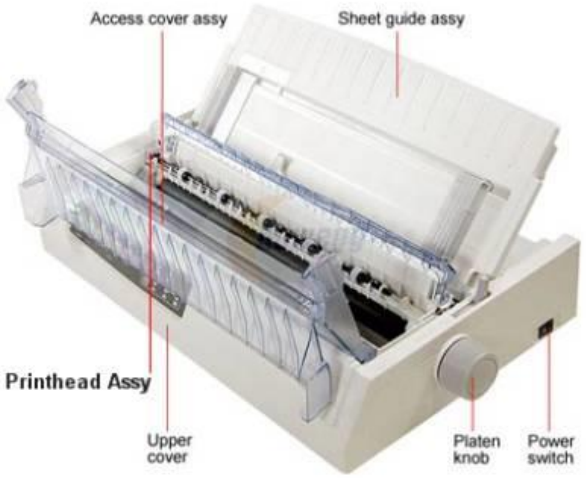
Disadvantages: noisy and cannot print graphics.

2.1.2 Dot-Matrix Printers

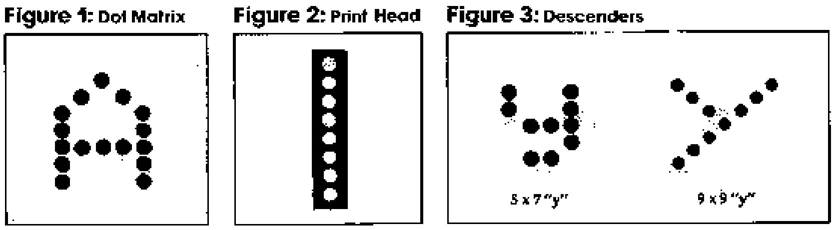
Dot-Matrix Printers use a column of pins on a moving printhead that strike an ink ribbon to form characters as dots. Most recent dot-matrix printers are equipped with 24 pins; they can print on multi-part stationery and continuous paper feed.

Advantages: lower running cost, rugged for dusty environments, negligible operator training;

Disadvantages: noisy, lower print quality, slow, unsuitable for photo printing.



Dot-matrix printer



8-pin dot-matrix print head with examples of printed letters

2.2 Solid-Ink Printers

Solid-Ink printers use ink in a waxy solid form that is melted and sprayed onto paper, avoiding liquid spillage.

Advantages: good print quality, ease of use, less waste, insensitive to paper thickness;

Disadvantages: higher power consumption, wax odor.

2.3 Thermal Printers

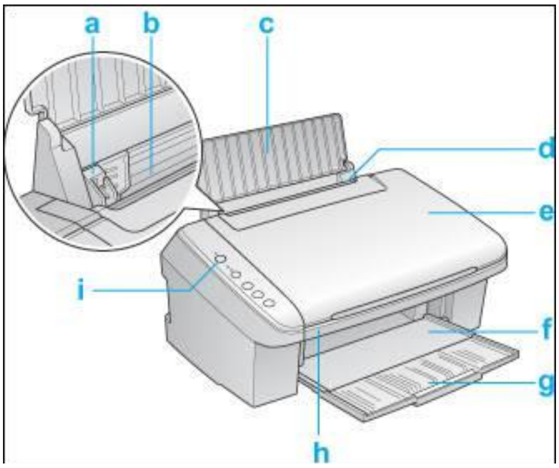
Thermal printers produce images by heating either thermochromic paper (direct thermal) or a heat-sensitive ribbon (thermal wax-transfer). Direct Thermal printers darken heat-sensitive paper, which can fade over time; Thermal wax-transfer printers melt wax-based ink from a ribbon, with speed limited by printhead heating/cooling.

2.4 Inkjet Printers

Inkjet printers spray extremely small droplets of aqueous ink through nozzles to create photo-quality images on paper. They use either thermal bubble or piezoelectric mechanisms to eject ink, and comprise parts such as edge guide, sheet feeder, print head assembly, and paper feed assembly.

Advantages: low cost, high quality, no warm-up time, quiet;

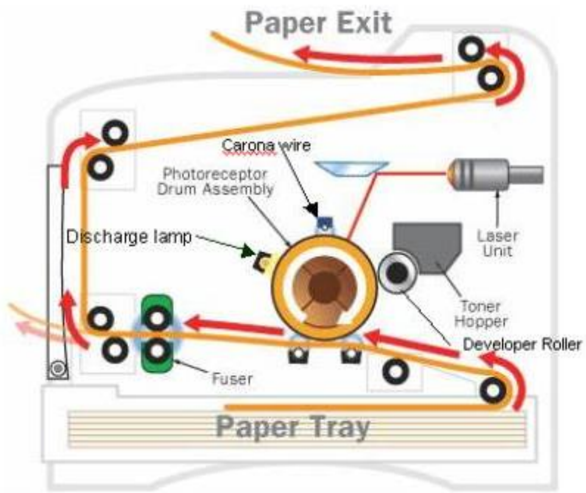
Disadvantages: expensive ink cartridges.

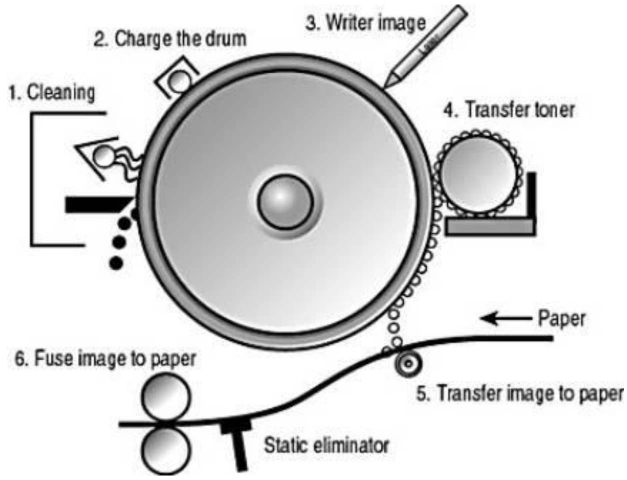


a typical ink jet printer

2.5 Laser Printers ⋆⋆

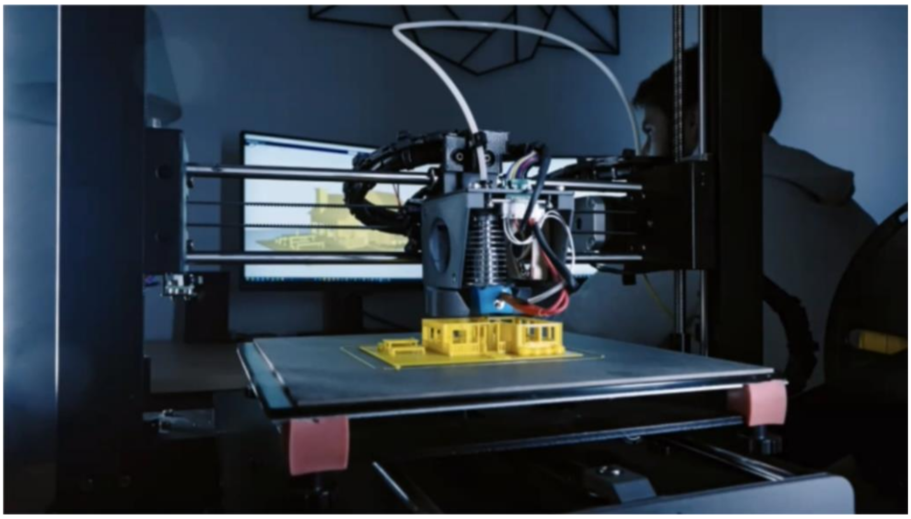
Laser printers use a laser beam and static electricity (electrophotostatic process) to form images on a photosensitive drum and transfer toner to paper. They are the fastest and most popular printers, producing extremely high-quality output near photo quality.





Laser printer components and their functionality

3D Printers ⋆⋆



3D Printing Uses:

- Construction
- Automotive
- PPE
- Medical Equipment
- Food
- Aerospace
- Education

The 3D Printing Process

3D printers are related to additive manufacturing. 3D printers use computer-aided design to understand a design. When a design is ready, a material that can be dispensed through a hot nozzle or precision tool is printed layer by layer to create a three-dimensional object from scratch.

When the modeling and slicing of a 3D object is completed, it's time for the 3D printer to finally take over. The printer acts generally the same as a traditional inkjet printer in the direct 3D printing process, where a nozzle moves back and forth while dispensing a wax or plastic-like polymer layer-by-layer, waiting for that layer to dry, then adding the next level. It essentially adds hundreds or thousands of 2D prints on top of one another to make a three-dimensional object.

How much 3D Printers cost?

– The cost of 3D printers vary based on the size, specialty and use. The cheapest 3D printers, for entry level hobbyists, typically range from $100 to $500. More advanced models can range between $300 and $5,000. Industrial 3D printers can cost up to $100,000.

Advantages

- Affordable
- Fast
- Work with Specialty Materials

Disadvantages

- May Not Provide Enough Strength
- May Have Accuracy Issues
- May Require Post-Processing

3. Introduction to Scanners ⋆⋆

A scanner is an input device that converts physical images, printed text, and handwriting into digital form by scanning one line at a time as the scanning head moves down the page. The main internal components of a scanner are the Glass Plate and Cover, the Scanning Head, and the Stepper Motor. Three common scanner form factors covered are Flatbed Scanners, Sheetfed Scanners, and Handheld Scanners.

Components inside a scanner are the following

i. Glass Plate and Cover

The glass plate is the transparent surface on which the original document is placed for scanning. The cover shields the document from stray light that could affect scan accuracy.

ii. Scanning head

The scanning head carries out the actual capture of the image. Its subcomponents include:

1. **Light source and mirror**: A bright white light illuminates the original and the reflected light is guided by mirrors towards the sensor.

2. **Stabilizer bar**: A stainless-steel rod that ensures smooth, vibration-free movement of the scanning head.
3. **CCD (Charge Coupled Device) or CIS (Contact Image Sensor)**: A CCD array converts incoming photons into electrical signals using lenses, while CIS uses a linear array of LEDs directly beneath the glass plate.

iii. Stepper Motor

A stepper motor moves the scanning head precisely down the page during the scan cycle, often via a belt drive.

3.1 Flatbed Scanners

The most commonly used scanner type, also called a desktop scanner, features a stationary glass plate on which the document rests while the scan head moves beneath it. Wide-format flatbed scanners can handle large maps or toposheets for high-quality large-area scans.

A Flatbed Scanner

3.3 Sheetfed Scanners

Sheetfed scanners feed individual sheets past a fixed scanning head, similar to a fax machine. They are limited to loose paper and can produce distorted images if the document is misaligned during feeding.

A Sheetfed Scanners

3.4 Handheld Scanners

Handheld scanners are compact and portable but can only scan up to about four inches in width. They require a steady hand to avoid distortion and are best suited for small logos or signatures rather than large documents or photographs.

A Handheld Scanners

Printer and Scanner Interfaces

A printer's interface is a combination of hardware and software that allows the printer to communicate with a computer. The hardware component—called a port—provides the physical connection, while the interface software manages data exchange. Every printer includes at least one interface, which incorporates both the communication type and requisite software.

There are eight major communication types:

1. Serial

Computers send one bit at a time over a serial connection; communication parameters such as parity and baud rate must be configured on both ends before data transfer begins.

2. Parallel

Parallel communication is faster than serial because it transmits eight bits simultaneously over eight separate wires. On PCs it uses a DB-25 connector, while printers employ a 36-pin connector.

3. USB (Universal Serial Bus)

USB interfaces automatically recognize new devices and can transfer data at up to 12 Mbps, making them a common choice for modern printers and scanners.

4. Network

Often implemented via Ethernet, network interfaces (NIC plus ROM-based firmware) enable laser printers and other devices to communicate with networks, servers, and workstations.

5. Infrared

Infrared interfaces use IR radiation to send print commands wirelessly between devices (e.g., laptops, PDAs, cameras) and printers, requiring an infrared acceptor on the printer.

6. SCSI (Small Computer System Interface)

SCSI connections—used by some laser and dye-sublimation printers—support daisy chaining multiple devices on a single bus and are relatively easy to implement.

7. IEEE 1394 Firewire

FireWire provides high-speed connectivity (up to 800 Mbps, and in later revisions up to 3.2 Gbps) and is typically used for digital video editing and other bandwidth-intensive applications.

8. Wireless

Wireless interfaces (infrared, Bluetooth, 802.11 Wi-Fi, etc.) transmit data through radio waves. Bluetooth in particular replaces cables over short distances (around 10 m) between computers and peripherals.

Out of these communication types, scanners most commonly use USB, Parallel, SCSI, and IEEE 1394/FireWire.

5. Installing Printers and Scanners ★★

Installing And Configuring Printers And Scanners:

- Step 1 : Attach the device using a local or network port and connect the power
- Step 2 : Install and Update the Device driver and calibrate the device
- Step 3 : Configure options and default settings
- Step 4 : Print/scan a test page

6. Printer and Scanner Problems and Troubleshooting

Laser printer problems (just main points for exam)

- **Print results too light, white/light columns**
  Possible cause: Low toner or a defective transfer corona
  Solution: Replace the toner cartridge

- **Ghosting**
  Description: Light images of previously printed pages appear on the current page
  Possible cause: Drum not fully discharged or erase lamp not functioning
  Solution: Print a few full-page black sheets or replace the cartridge and erase lamp

- **Prints black pages**
  Possible cause: Damaged primary corona
  Solution: Replace the primary corona

- **Memory Overflow error**
  Possible cause: Image to be printed is too large for printer memory
  Solution: Reduce print resolution or add more RAM to the printer

- **Paper jams**
  Possible causes: Worn paper-feed rollers or broken pickup-roller drive gear
  Solutions: Replace worn rollers or the damaged drive gear

- **Repetitive small marks or defects**
  Possible causes: Toner spillage inside the printer or a cracked drum roller accumulating toner
  Solution: Replace the offending roller

- **Image Smudging**
  Description: Printed image smears when touched
  Possible cause: Fuser not heating properly
  Solution: Replace the fuser assembly

Dot-Matrix printer problems (just main points for exam)

- **Print results are too light**
  Possible cause: Ink ribbon cartridge is out of ink
  Solution: Replace the ink ribbon

- **Printer does not print**
  Possible causes: Printer offline, out of paper, or cable not connected
  Solutions: Bring printer online, load paper, or check cable connections

- **Characters are incomplete**
  Possible cause: Struck or broken pins in the print head
  Solution: Replace the print head

- **Printout jams inside the printer**
  Possible causes: Obstruction in the paper path or stripped drive gears
  Solutions: Disassemble to remove blockage or replace the stripped gears

- **Stepper motor problems**
  Description: Print-head movement or paper advance fails when motor is forced
  Solution: Replace the damaged stepper motor

Scanner problems and troubleshooting

Common scanner issues include power-on failures, unusual noises, or failure to scan. To troubleshoot:

- Ensure scanner software is correctly installed and configured
- Confirm the scanner's power light is on
- Uninstall and reinstall scanner software if errors occur

- Run manufacturer-provided diagnostic tests
- Verify that cables are securely connected
- Protect the scanner from overheating, dust, and humidity
- Restart the computer, unplug/replug cables, and check for hardware conflicts in Device Manager

7. Tools to Troubleshooting Printer/Scanner Problems

Basic troubleshooting tools include:

- **Multimeter**
- **Screw driver**
- **Cleaning solutions**
- **Extension Magnets**
- **Test patterns**

References

1. https://www.tutorialsweb.com/computers/printers-scanners/.htm
2. https://www.techexams.net
3. https://www.howstuffworks.com
4. https://www.brainbell.com